

ZK Proofs as Money: A New Economic Primitive for Blockchain-Native Value

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Builds on: Ephemeral Quorum Subnets and Quorum Attestation (Guberman, May 2, 2026, aprilgatehq.com/whitepaper)

ABSTRACT

Every blockchain that has attempted to define "useful work" has either been too narrow — limiting valid work to one specific computation — or has collapsed into gaming when the definition was stretched. This paper argues that the missing ingredient is an economic filter, not a governance mechanism. We introduce a two-category legitimacy test: work is legitimate if it has intrinsic commodity value (universally hard and useful, independent of any specific payer) or demonstrated monetary value (an external party pays market price for the output). Work satisfying neither test is speculation and earns no consensus weight. Within this framework, zero-knowledge proof generation occupies a unique position: it satisfies both categories simultaneously, making it the natural unit of account — the money — of a blockchain economy built on Ephemeral Quorum Subnets. AI inference and hard computation become commodities that back the currency. Paid useful work — cold chain compliance, federated medical AI, algorithmic audit — becomes traded goods whose value is set by external markets, not internal protocol design. The result is a self-regulating economy that needs no governance committee to decide what counts as useful: the market does.

Inspiration and Acknowledgment

This paper would not exist without two contributions from David Lieberman and the Gonka.ai team. First, the Gonka whitepaper's articulation of AI inference as the unanimous definition of useful blockchain computation — the cleanest and most principled existing answer to the useful-work problem — provided the foundation this paper builds on and generalizes. The discipline of keeping that definition narrow and honest is itself a theoretical contribution that this paper takes seriously. Second, a conversation in which David described attempts to count social media engagement as useful work — and why that failed — provided the negative example that made the positive criterion clear. The two-category legitimacy test in Section 3 is a direct response to both. The conclusions are the author's

own; the inspiration is genuinely shared.

1. The Useful-Work Problem

Bitcoin's Proof-of-Work solved one problem elegantly: making attack expensive. It solved it by making the work itself expensive and intentionally useless. Energy burned on arbitrary hashing produces nothing of value to anyone outside the network. That is a feature, not a bug — it means the only reason to do the work is to participate honestly in consensus.

The obvious next question: can the work be useful to the world AND serve as proof of honest participation? If so, you get security for free as a byproduct of genuinely valuable computation. This is the promise that has motivated a decade of "useful work" blockchain research.

The attempts have been instructive. AI inference is hard to fake, economically valuable, and verifiable — a strong definition. Social media engagement is easy to game, economically hollow when manufactured at scale, and impossible to verify as genuine — the definition collapsed immediately. Scientific computation sits somewhere between: hard to fake, genuinely valuable, but verification requires re-execution or trust.

The pattern across failures is consistent: the protocol tried to *define* useful work by enumeration, and adversaries found the gaps. The pattern across successes is equally consistent: the work had an external economic anchor that made gaming it more expensive than the reward. AI inference works not because the protocol says it works, but because real clients pay real money for real inference, and fabricating that output is detectable. The external demand is the immune system. The protocol definition is just the label.

2. The Missing Filter

The question is not "what work should the protocol declare valid?" That question leads to governance, committees, and capture. The question is "what property must work have to be self-evidently legitimate, without anyone needing to decide?"

Consider how commodity markets answer this. Gold, oil, wheat, and natural gas are not money — but they are so undeniably useful, so costly to produce, and so universally demanded that markets treat them as stores of value without any authority declaring them so. Their legitimacy is self-evident from their physical properties and their demand. Nobody voted gold into being valuable.

The same principle can be applied to blockchain computation. Work is legitimate if it meets at least one of two tests:

- **Intrinsic commodity value:** the work is universally recognized as hard, costly, and useful independent of any specific transaction. Its value does not depend on a particular

payer agreeing. Running a transformer correctly, generating a ZK proof, folding a protein — these have value the way oil has value: physically real, universally demanded, impossible to fake at scale.

- **Demonstrated monetary value:** an external party pays market price for the work output. The payment is the proof of demand. A pharma insurer paying for cold chain compliance attestation, a hospital consortium paying for federated model training, a regulator paying for an algorithmic audit — these establish value through market transaction, not protocol declaration.

Work satisfying neither test — manufactured social media engagement, vanity hashes, internal gaming — earns nothing. Not because a committee decided it shouldn't, but because no external anchor gives it value. The market is the filter. The protocol is just the verification layer.

3. The Three-Layer Economy

Applying the two-category test to the full range of blockchain-useful computation produces a natural three-layer economic structure that mirrors the structure of any mature economy: money, commodities, and traded goods.

3.1 ZK Proof Generation — Money

Zero-knowledge proof generation occupies a unique position: it satisfies both legitimacy tests simultaneously.

- **Intrinsic commodity value:** generating a valid ZK proof is computationally expensive (GPU-hours for large circuits), mathematically hard to fake (proof soundness is a cryptographic guarantee, not a statistical one), and universally useful (every Ephemeral Quorum Subnet produces a Quorum Attestation that requires ZK proof generation). The demand for ZK proofs is structural, not optional.
- **Demonstrated monetary value:** upstream clients — pharma insurers, hospital consortia, research institutions, financial regulators — pay for ZK proofs because the proofs fulfill real compliance, audit, and integrity obligations. The payment flow is external and market-priced.

The self-verifying property of ZK proofs makes them uniquely suited as the unit of account. A ZK proof either verifies or it does not. There is no sampling, no trust, no probabilistic estimate. The proof IS the verification. This makes ZK proof generation the only category of blockchain work that is simultaneously: hard to fake, trivially verified, externally demanded, and structurally required by the network's own operation. It is, in the most precise sense, the currency the chain runs on.

3.2 Hard Computation — Commodity

AI inference, ML training, scientific simulation, and cryptographic computation satisfy the intrinsic commodity value test. They are hard to fake, universally useful, and costly to produce.

Their value does not depend on any specific payer — the world needs more inference, better models, and more simulation regardless of who is paying today.

These workloads are the gold and oil of this economy. They back the currency — ZK proofs are generated to attest that this computation happened correctly — and they have intrinsic value that makes them resistant to gaming. You cannot manufacture fake AI inference at lower cost than real inference without the fabrication being detectable. The physical cost of the computation is the economic floor.

Crucially, the same GPU hardware that runs AI inference is well-aligned with ZK proof generation — both are dominated by matrix operations over finite fields. A host node doing inference and a host node generating ZK proofs for upstream sensor networks are using the same silicon more fully. The hardware does not distinguish between the two workloads. This alignment is not coincidental: it is the physical basis for the economic unification this paper proposes.

3.3 Paid Useful Work — Traded Goods

A broader category of work satisfies the demonstrated monetary value test without necessarily having intrinsic commodity value. Temperature sensor attestation for a pharma shipment is not universally demanded the way AI inference is — but a specific insurer is paying specific money for a specific proof, and that payment establishes the value unambiguously.

Four instantiations illustrate the range:

- **Pharmaceutical cold chain (April Gate):** hardware-attested sensor nodes attest temperature compliance during shipment. The pharma insurer pays for the ZK proof. The sensor operator earns revenue. The host node generating the ZK proof earns both client payment and consensus weight. The work is legitimate because someone with real money validated its value.
- **Federated medical AI:** a hospital consortium trains a shared diagnostic model without any hospital exposing patient data. The consortium pays for the ZK proof of correct federated training. No central server. No data breach surface. The payment establishes the value; the ZK proof establishes the integrity.
- **Distributed scientific computation:** research institutions pay for provably correct simulation results. The ZK proof gives the finding cryptographic provenance without requiring trust in any central institution. Grant funding is the external demand signal.
- **Algorithmic trading audit:** financial regulators require proof that a trading algorithm behaved within regulatory bounds. The regulated firm pays for the audit proof. The regulator gets a binding result. The trader retains IP. Regulatory obligation is the external demand signal.

In each case the pattern is identical: external party pays → subnet witnesses and attests → ZK proof generated → QA posted to L1 → host earns client payment plus consensus weight. The

market price of the work, not the protocol, determines its value.

4. The Economic Immune System

The two-category test creates a self-regulating economy without requiring any governance mechanism to define valid work. Consider what happens to an actor trying to game the system:

- **Fake ZK proofs:** cryptographically impossible. Proof soundness guarantees a valid proof cannot be constructed without performing the underlying computation. The attack surface does not exist.
- **Manufacturing fake demand:** an actor could pay themselves to generate proofs for worthless computations. But the cost of generating the proof exceeds the consensus weight earned, because the weight is calibrated to the computational complexity of the circuit — not the economic value of the underlying task. Gaming this requires spending more than you earn. The economic floor is the computation cost itself.
- **Claiming intrinsic value for worthless work:** social media engagement, vanity hashes, internal coordination tasks — these fail the intrinsic commodity value test because no external market treats them as stores of value. The protocol does not need to ban them; they simply produce no consensus weight because no legitimate ZK proof of useful work can be constructed for them.

The immune system is economic, not political. It does not require trust in the Workload Type Registry or any governance body to make correct decisions. It requires only that proof generation is hard, verification is cheap, and external demand is real. All three of these are properties of the cryptography and the market, not properties of the protocol.

"If the cost of doing the work exceeds the reward of the stake, nobody does it unless someone is actually paying for the work. The payment is the proof of demand. The cost is the DDOS deterrent."

5. Connection to Ephemeral Quorum Subnets

This paper's economic primitive and the architectural primitive developed in the companion paper — Ephemeral Quorum Subnets and Quorum Attestation (Guberman, May 2, 2026) — are two sides of the same design. The architectural paper asks: what is the minimal L1 that can support heterogeneous subnet computation? This paper asks: what is the economic model that makes that L1 self-sustaining and self-regulating?

The answers interlock precisely:

- Every EQS subnet produces a Quorum Attestation.
- Every QA requires ZK proof generation by a host node.
- ZK proof generation is the currency the L1 runs on.

- The work that secures the chain IS the work the chain is selling.

There is no separation between the security mechanism and the revenue mechanism. This is the property that neither Bitcoin (security mechanism produces nothing sellable) nor current AI compute networks (revenue mechanism is probabilistically verified, not cryptographically guaranteed) have achieved.

The internet analogy from the companion paper applies here too. IP did not define what valid traffic looks like. Applications that people were willing to pay for thrived. Applications nobody valued died. The network did not adjudicate — economics did. This paper proposes the same model for blockchain useful work: the L1 verifies proofs and settles payments. The market decides what work is worth doing.

6. Implications for Network Design

If ZK proof generation is the baseline currency, several design implications follow naturally:

- **Host node specialization is healthy.** Some nodes optimize for AI inference throughput. Others optimize for ZK proof generation (different hardware profile — fast memory bandwidth, specific CPU characteristics alongside GPU parallelism). The network accommodates both because both produce legitimate currency. When AI inference demand is low, ZK proof demand from upstream subnets absorbs spare capacity. Revenue is more stable across demand cycles.
- **The Workload Type Registry becomes a compatibility list, not a permission list.** It specifies how to verify each workload type — not whether that workload is allowed. Any work that produces a verifiable ZK proof and has external demand is implicitly valid. The registry is updated to support new proof formats as cryptographic tooling matures, not to grant or revoke permission to specific industries.
- **Decentralization is preserved at L1.** Because ZK verification is $O(1)$ regardless of computation complexity, any node can verify any proof. The L1 validator set does not need to understand pharmaceutical compliance, medical AI, or financial regulation. It needs only to verify proofs. This keeps the validator set large, diverse, and decentralized — exactly the property the trilemma says is impossible to maintain with scalability. The EQS architecture makes it possible by keeping L1 minimal.
- **The gap in the proof sequence is an auditable event.** A cold chain shipment that produces no QA during a time window is not a system failure — it is a tamper signal. An AI compute node that stops producing verified work is not offline — it is a potential defector. The absence of proof is itself information, and that information is permanently recorded by its absence from the L1 sequence.

7. Open Questions

- **Calibrating consensus weight to circuit complexity.** If weight is proportional to constraint count, large circuits earn disproportionate weight. The right calibration function — linear, logarithmic, capped — is an open mechanism design problem. The wrong choice creates incentives to generate unnecessarily large circuits.
- **Handling work with intrinsic but unverifiable value.** Some genuinely useful computation — certain types of scientific simulation — is not easily expressible as an arithmetic circuit and therefore not currently provable via standard ZK systems. As zkVM and general-purpose ZK tooling mature, this boundary will shift. The framework accommodates this evolution: as new work becomes provable, it becomes currency.
- **The transition from probabilistic to cryptographic verification.** Networks currently using probabilistic sampling for AI inference verification face a transition problem: moving to ZK verification requires prover infrastructure that does not exist at scale today. The proof generation cost for large transformer inference is currently prohibitive. This is an engineering problem with a clear trajectory, not a theoretical barrier.
- **Cross-network QA recognition.** For the economy described here to reach its potential, multiple L1 chains must recognize each other's QA formats. A cold chain proof generated for Solana should be verifiable on Ethereum and vice versa. This requires standardization that is as much a coordination problem as a technical one.

8. Conclusion

The useful-work problem in blockchain has resisted solution because every attempted solution has tried to answer the wrong question: "what work should we declare valid?" The right question is "what property makes work self-evidently legitimate without anyone needing to decide?"

The two-category legitimacy test — intrinsic commodity value or demonstrated monetary value — answers that question without governance, enumeration, or trust in any authority. ZK proof generation satisfies both categories simultaneously, making it the natural currency of a blockchain economy built on ephemeral subnet computation. Hard computation becomes the commodity that backs the currency. Paid useful work becomes the traded goods whose value the market sets.

The result is a blockchain economy that is self-regulating by design: the security mechanism and the revenue mechanism are the same mechanism. The work that secures the chain is the work the chain is selling. The market filters invalid work without requiring the protocol to enumerate valid work. And the absence of proof is itself an auditable event — making the chain not just a ledger of what happened, but a ledger of what was proven to have happened.

April Gate is the first instantiation of this economy at the smallest possible scale: four sensor nodes, one subnet, one proof, one insurer. The architecture scales from there.

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